

HW2

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Problem 1

(a)

We are given the update rule with momentum as:

$$w_{t+1} = w_t - \eta z_{t+1} \quad (1)$$

$$z_{t+1} = (1 - \beta)z_t + \beta(2X^T X w_t - 2X^T y) \quad (8)$$

We are to reparameterize the vectors using the SVD of the data matrix $X = U\Sigma V^T$, with singular values $\sigma_1, \dots, \sigma_m$. Let $x_t = V^T(w_t - w^*)$, and $a_t = V^T z_t$. That is:

$$x_t = V^T(w_t - w^*) \quad (2)$$

$$a_t = V^T z_t \quad (9)$$

We will now rewrite Eq. (8) using $x_t[i]$ and $a_t[i]$. First, note that since $X = U\Sigma V^T$, then:

$$X^T X = V\Sigma^T U^T U \Sigma V^T = V\Sigma^2 V^T$$

So,

$$X^T X w_t = V\Sigma^2 V^T w_t \quad \text{and} \quad X^T y = X^T X w^* = V\Sigma^2 V^T w^*$$

Substitute into Eq. (8):

$$z_{t+1} = (1-\beta)z_t + \beta(2X^T X w_t - 2X^T y) = (1-\beta)z_t + \beta(2V\Sigma^2 V^T w_t - 2V\Sigma^2 V^T w^*)$$

Factor out $V\Sigma^2 V^T$:

$$z_{t+1} = (1 - \beta)z_t + \beta \cdot 2V\Sigma^2 V^T(w_t - w^*)$$

Now multiply both sides by V^T :

$$V^T z_{t+1} = (1 - \beta)V^T z_t + \beta \cdot 2\Sigma^2 V^T (w_t - w^*) \Rightarrow a_{t+1} = (1 - \beta)a_t + 2\beta\Sigma^2 x_t$$

Similarly, using $w_{t+1} = w_t - \eta z_{t+1}$, we get:

$$w_{t+1} - w^* = w_t - w^* - \eta z_{t+1}$$

Apply V^T to both sides:

$$x_{t+1} = x_t - \eta a_{t+1}$$

Solutions are

$$x_{t+1} = x_t - \eta a_{t+1} \tag{3}$$

$$a_{t+1} = (1 - \beta)a_t + 2\beta\Sigma^2 x_t \tag{4}$$

(b) From Eq. (9), we reparameterize the variables:

$$x_t = V^T (w_t - w^*), \quad a_t = V^T z_t.$$

Using the SVD of $X = U\Sigma V^T$, with $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_m)$, we rewrite Eq. (8) using $x_t[i]$ and $a_t[i]$:

$$\begin{aligned} x_{t+1}[i] &= x_t[i] - \eta a_{t+1}[i], \\ a_{t+1}[i] &= (1 - \beta)a_t[i] + \beta(2\sigma_i^2 x_t[i]). \end{aligned}$$

(c) The recurrence can be written in matrix form:

$$\begin{bmatrix} a_{t+1}[i] \\ x_{t+1}[i] \end{bmatrix} = R_i \begin{bmatrix} a_t[i] \\ x_t[i] \end{bmatrix},$$

where

$$R_i = \begin{bmatrix} 1 - \beta & 2\beta\sigma_i^2 \\ -\eta(1 - \beta) & 1 - 2\eta\beta\sigma_i^2 \end{bmatrix}.$$

To find the eigenvalues of R_i , consider the characteristic equation:

$$\lambda^2 - \text{tr}(R_i)\lambda + \det(R_i) = 0,$$

where

$$\text{tr}(R_i) = (1 - \beta) + (1 - 2\eta\beta\sigma_i^2), \quad \det(R_i) = (1 - \beta)(1 - 2\eta\beta\sigma_i^2) + 2\eta\beta\sigma_i^2(1 - \beta).$$

Simplifying,

$$\text{tr}(R_i) = 2 - \beta - 2\eta\beta\sigma_i^2, \quad \det(R_i) = (1 - \beta).$$

Eigenvalues are:

$$\lambda = \frac{1}{2} \left(\text{tr}(R_i) \pm \sqrt{\text{tr}(R_i)^2 - 4\det(R_i)} \right).$$

(d) When eigenvalues are repeated, the discriminant is zero:

$$(2 - \beta - 2\eta\beta\sigma_i^2)^2 = 4(1 - \beta).$$

Solving this equation for η gives the condition for repeated eigenvalues.

For them to be real and repeated *and* inside the unit circle, the absolute value of the eigenvalue must be < 1 :

$$|\lambda| = \left| \frac{2 - \beta - 2\eta\beta\sigma_i^2}{2} \right| < 1.$$

The highest η with repeated, stable eigenvalues is found by saturating this inequality and ensuring $\eta > 0$.

(e) For distinct real eigenvalues to be inside the unit circle:

$$|\lambda_1| < 1, \quad |\lambda_2| < 1.$$

This gives a bound on η in terms of β and σ_i . The range of η can be found by numerically solving the inequality involving λ 's from the quadratic above.

(f) For complex eigenvalues, the discriminant must be negative:

$$(2 - \beta - 2\eta\beta\sigma_i^2)^2 < 4(1 - \beta).$$

The eigenvalues $\lambda = re^{\pm i\theta}$ are stable iff $r = \sqrt{\det(R_i)} = \sqrt{1 - \beta} < 1$, which always holds since $0 < \beta < 1$.

To maximize learning rate η , we solve:

$$(2 - \beta - 2\eta\beta\sigma_i^2)^2 = 4(1 - \beta)$$

and isolate η :

$$\eta = \frac{2 - \beta \pm \sqrt{4(1 - \beta)}}{2\beta\sigma_i^2}.$$

We choose the largest such η consistent with stability.

(g) Given: $\beta = 0.1$, $\sigma_{\max}^2 = 5$, $\sigma_{\min}^2 = 0.05$.

For normal gradient descent (no momentum), optimal rate is:

$$\text{rate}_{\text{GD}} = \frac{(\sigma_{\max}/\sigma_{\min})^2 - 1}{(\sigma_{\max}/\sigma_{\min})^2 + 1} = \frac{(5/0.05)^2 - 1}{(5/0.05)^2 + 1} = \frac{10000 - 1}{10000 + 1} \approx 0.9998.$$

This is very slow convergence. To reach within 0.1

$$\text{rate}^T = 0.001 \Rightarrow T = \frac{\log 0.001}{\log \text{rate}} \approx \frac{-6.91}{\log 0.9998} \approx 34500 \text{ iterations.}$$

With momentum and optimal η for complex roots, the effective convergence factor is $\sqrt{1 - \beta} \approx \sqrt{0.9} \approx 0.9487$.

$$T_{\text{momentum}} \approx \frac{\log 0.001}{\log 0.9487} \approx \frac{-6.91}{\log 0.9487} \approx 131.$$

So using momentum can reduce iteration count drastically, from 34500 to 131, to achieve the same level of convergence.

Problem 2

(a)

Algorithm 2 Adam Optimizer (without bias correction)

1. Given $\eta = 0.001, \beta_1 = 0.9, \beta_2 = 0.999$
2. Initialize: time step $t \leftarrow 0$, parameter $\theta_t = 0 \in \mathbb{R}^n$, $m_t \leftarrow 0$, $v_t \leftarrow 0$
3. Repeat
4. $t \leftarrow t + 1$
5. $g_t \leftarrow \nabla f_t(\theta_{t-1})$
6. $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$ (A)
7. $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ (B)
8. $\theta_t \leftarrow \theta_{t-1} - \eta \frac{m_t}{\sqrt{v_t} + \epsilon}$ (where ϵ is a small constant for numerical stability, often 10^{-8})
9. Until the stopping condition is met

(b)

We consider the update rule for regular SGD on the L2-regularized loss $f_t^{reg}(\theta)$. The update rule for SGD is:

$$\theta_t = \theta_{t-1} - \eta \nabla f_t^{reg}(\theta_{t-1})$$

First, let's find the gradient of the L2-regularized loss:

$$\begin{aligned} \nabla f_t^{reg}(\theta) &= \nabla \left(f_t(\theta) + \frac{\lambda}{2} \|\theta\|^2 \right) \\ \nabla f_t^{reg}(\theta) &= \nabla f_t(\theta) + \nabla \left(\frac{\lambda}{2} \sum_{i=1}^n \theta_i^2 \right) \\ \nabla f_t^{reg}(\theta) &= \nabla f_t(\theta) + \frac{\lambda}{2} \cdot 2\theta \\ \nabla f_t^{reg}(\theta) &= \nabla f_t(\theta) + \lambda\theta \end{aligned}$$

Now, substitute this gradient back into the SGD update rule for $f_t^{reg}(\theta)$:

$$\begin{aligned} \theta_t &= \theta_{t-1} - \eta (\nabla f_t(\theta_{t-1}) + \lambda \theta_{t-1}) \\ \theta_t &= \theta_{t-1} - \eta \nabla f_t(\theta_{t-1}) - \eta \lambda \theta_{t-1} \\ \theta_t &= (1 - \eta \lambda) \theta_{t-1} - \eta \nabla f_t(\theta_{t-1}) \end{aligned}$$

Now, let's compare this with the weight decay update rule:

$$\theta_t = (1 - \gamma)\theta_{t-1} - \eta \nabla f_t(\theta_{t-1})$$

By comparing the two equations, we can see that for the two update rules to be equivalent, we must have:

$$1 - \gamma = 1 - \eta\lambda$$

$$\gamma = \eta\lambda$$

Thus, SGD with weight decay using the original loss $f_t(\theta)$ is equivalent to regular SGD on the L2-regularized loss $f_t^{reg}(\theta)$ when γ is chosen as $\eta\lambda$.

Problem 4

(a)

The 1-layer neural network learns a smoothed approximation of the target function f_{true} . The features it learns are mostly global. The principal features are smooth but not expressive enough to model certain patterns.

(b)

```
0.weight 0.00293884
2.weight 0.00189422
4.weight 0.00091355
6.weight 0.00037678
8.weight 0.00020603
```

The gradient norms are extremely small, meaning that the network suffers from the vanishing gradients problem during training.

(c)

The model fails to fit the data well. The function it learns remains nearly flat due to low activation and minimal gradient updates, causing training to progress very slowly.

(d)

Before training, singular values decay rapidly, which is indicative of a low-rank gradient matrix. Principal features are flat as well. After training, the singular values increase slightly, yet the representation remains limited. The learned features stay mostly uniform, showing poor adaptation to the input.

(e)

```
0.weight 14.03246975
2.weight 12.88746166
4.weight 10.92749310
6.weight 9.08419227
8.weight 8.13409328
```

These values are quite large, which can cause instability during training due to the exploding gradients problem.

(f)

The model fails to converge and behaves erratically. It approximates the target function poorly, and training oscillates without reaching a low-loss region.

(g)

```
0.weight 0.29059160
2.weight 0.26891512
4.weight 0.22682744
6.weight 0.19063750
8.weight 0.16098435
```

The gradient norms are well-scaled.

(h)

Now, with proper initialization, singular values have smoother decay, and top principal features already reflect input structure. The large weight initialization shows noisy, unstructured features and a less meaningful SVD spectrum.

(i)

The model fits the target function well, learning the oscillatory structure. It converges faster than previous setups (and generalizes better), due to balanced activations and gradient propagation.

(j)

The shallow model captures the broad shape of the function but misses oscillations. Its features and singular values remain low-rank. The deep model has more significant singular values allowing it to generalize better.

(k)

The model fits the data well in the observed region but performs poorly for $x > 0.4$. The lack of training data in this region leads to poor generalization.

(1)

With mismatched training data, singular values and features drop in expressiveness for unobserved regions. The matched-data model has stronger, more diverse feature directions across the full domain, leading to better generalization of f_{true} .